SDR 184 Final Project: eSports Earning Analysis

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Introduction

eSports is a really interesting topic because of how global it is. Almost anyone in the world can compete and there's competition in almost every game imaginable, and when we saw that the eSports earnings website was one of the data sources available for this project, we were interested.

This data contains the eSports earning for players for the past 20+ years. It contains data from every major to mid sized eSport, from players from pretty much every country in the world. This data was chosen due to it being really rich and having a lot of different aspects to analyze. The way this data was obtained was from https://www.esportsearnings.com/, and all the data in this website was entered manually, getting it from forums, tournament organizers, and all made sure to be accurate.

This data follows the FAIR principles because this data is pretty findable since you can go to that website and find all the data that I used, its accessible since its free and anyone can use it for whatever they need, its interoperable because you can modify it and pretty easily, and it's reusable because the data will stay up and anyone can use it.

This follows the CARE principles because it is for collective benefit since it's cool and interesting and helps people understand how everyone can be good at esports, this data has authority to control because people are empowered by it, and its fair. It's responsible because it has no bias, and it accurately represents groups. It's ethical because it hurts no one and it makes sure to treat everyone equally.

The specific attributes of this data that im focused on, is specifically the geographical differences between players of different games, and what regions excel in different games. There's a lot of data on different players and where they're from, and this could make a really interesting analysis.

First Analysis

The first data analysis I decided to do, was to take the statistics of the top earning countries of all time and analyze it. I took the data from eSports earnings which gives the highest earning countries, with the amount of money, and how many players in each country. I was really curious what countries in general are the best at eSports since it could be anything. Tidying this in R gave me this data:

	Rank	Country	Total Earnings (USD)	Players	Earnings per Player (USD)
1	1	China	$333,124,854	9,387	$35,488
2	2	United States	$302,289,324	29,501	$10,247
3	3	Korea, Republic of	$157,154,935	5,987	$26,249
4	4	Russian Federation	$98,801,881	5,900	$16,746
5	5	Brazil	$74,212,435	6,058	$12,250
6	6	Denmark	$64,982,128	2,212	$29,377
7	7	France	$63,290,937	6,705	$9,439
8	8	Sweden	$59,203,302	3,367	$17,583
9	9	Germany	$53,321,255	6,987	$7,631
10	10	Canada	$50,919,207	4,395	$11,586
11	11	United Kingdom	$50,674,464	5,747	$8,818
12	12	Japan	$43,535,933	4,004	$10,873
13	13	Ukraine	$41,266,588	1,727	$23,895
14	14	Australia	$34,849,292	4,639	$7,512
15	15	Finland	$34,242,303	2,216	$15,452
16	16	Poland	$32,938,128	3,164	$10,410
17	17	Thailand	$31,614,841	2,319	$13,633
18	18	Philippines	$27,381,660	1,918	$14,276
19	19	Malaysia	$23,483,101	1,723	$13,629
20	20	Taiwan, Republic of China	$23,126,488	1,726	$13,399

Using the tidied data from this graph, I was able to create a visualization using a bar graph, which gave me this:

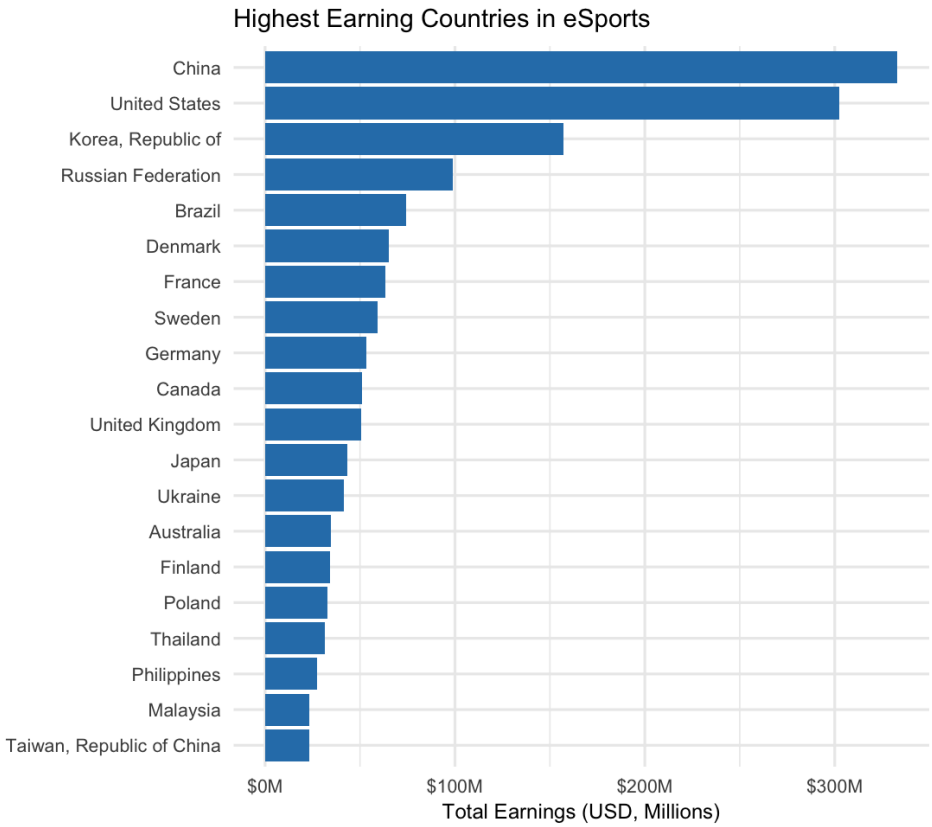


Figure 1. Total eSports earnings by country for the top 20 earning countries (USD, shown in millions).

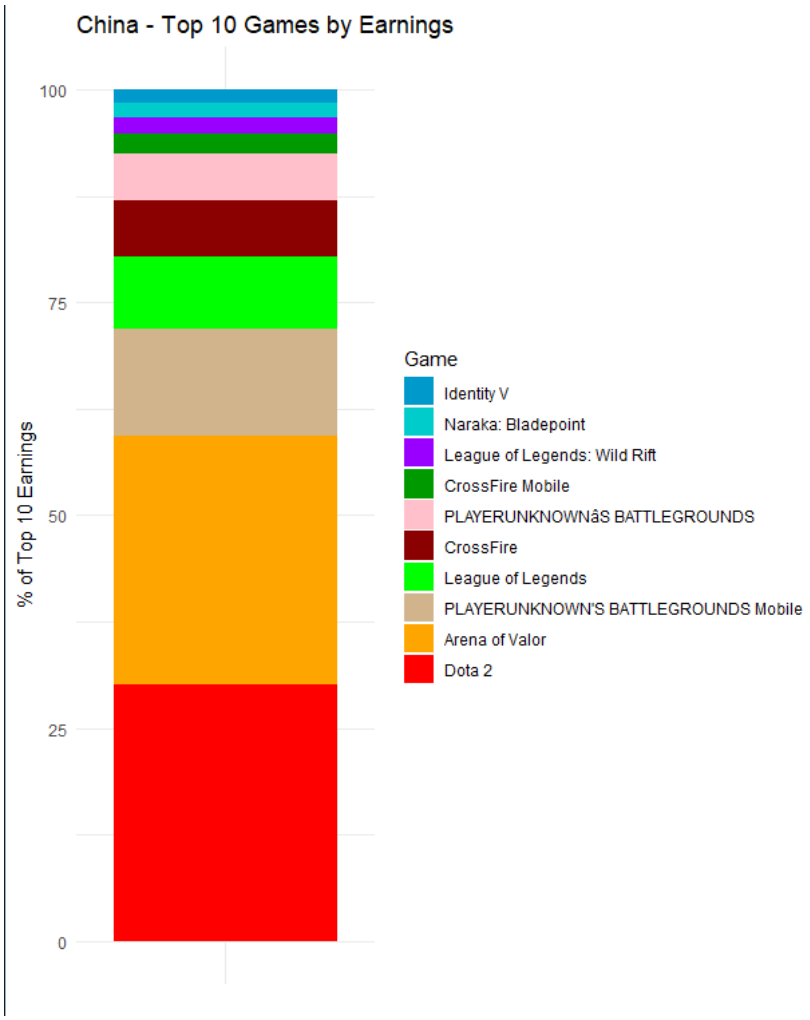
Alt text: This graph is a horizontal bar graph that demonstrates the highest earning countries in eSports. At the very top, we have China, United States Korea, Russia, and Brazil. Then, it decreases pretty linearly. China has earned around 350 millon USD, and has a pretty exponential decrease.

There's a few conclusions I've reached from this data. Firstly, a country having a bigger population has a direct correlation with how much success a country has in eSports. China, United States, Russia, And Brazil are some of the biggest countries in the world, and this makes sense because if there's more people in a country, there's going to be more people competing.

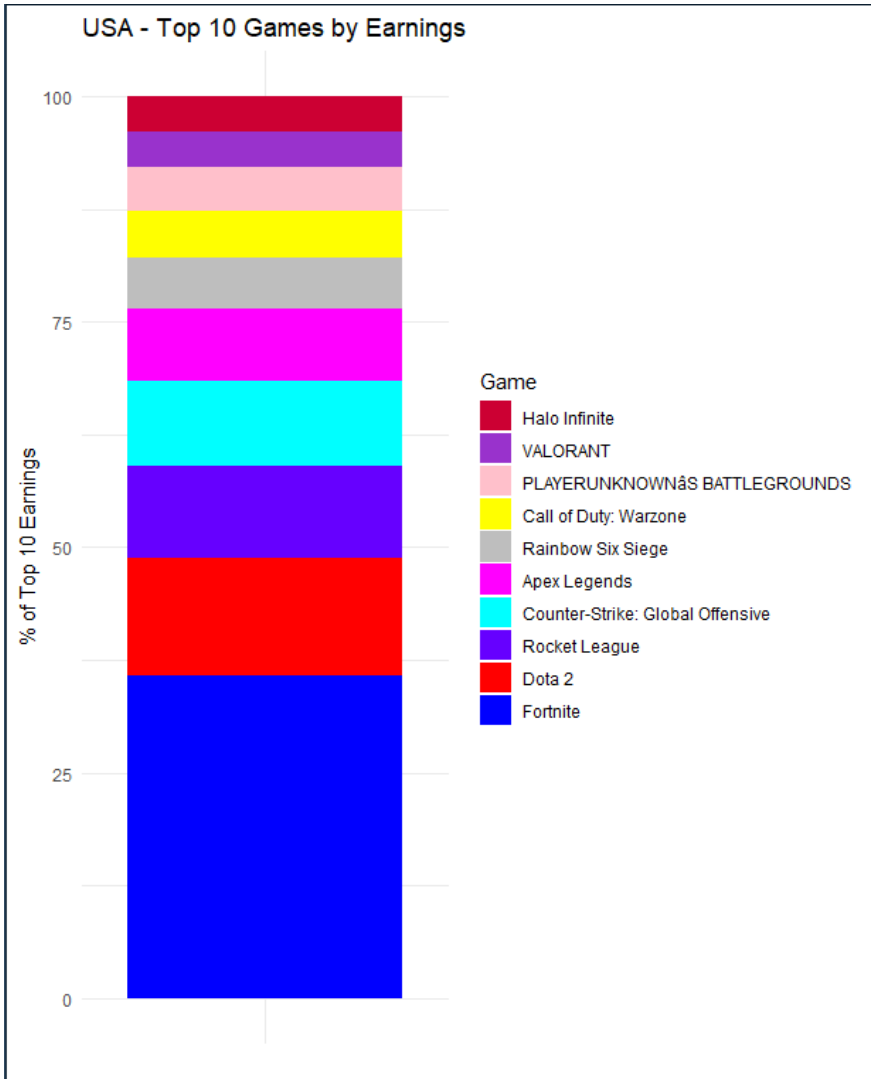
Second Analysis

The second analysis I've decided to do is a stacked bar chart of the top 10 games per country, looking specifically at the top 3 countries (China, United States, Korea). I took the data from eSportsearnings, which shows top games per country.

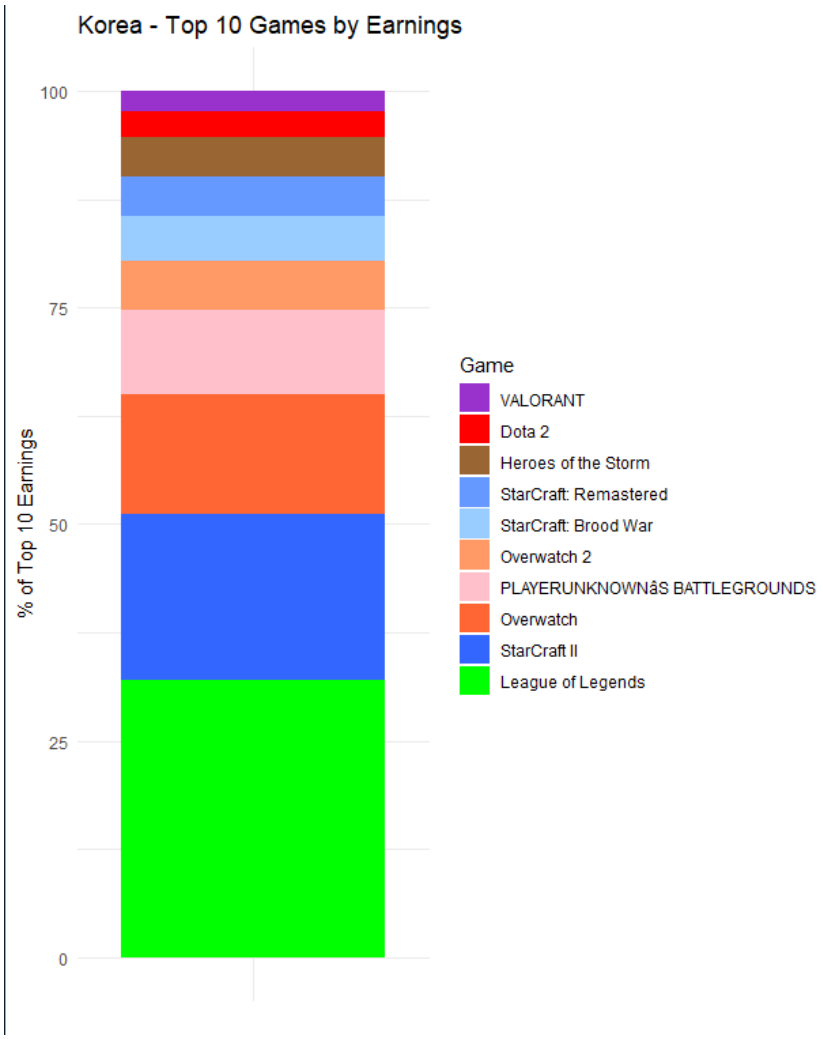
	Game	Pct
1	Dota 2	30.156010
2	Arena of Valor	29.155023
3	PLAYERUNKNOWN'S BATTLEGROUNDS Mobile	12.628404
4	League of Legends	8.355541
5	CrossFire	6.597197
6	PLAYERUNKNOWNâ□□S BATTLEGROUNDS	5.624853
7	CrossFire Mobile	2.353099
8	League of Legends: Wild Rift	1.809400
9	Naraka: Bladepoint	1.667873
10	Identity V	1.652600



	Game	Pct
1	Fortnite	35.762415
2	Dota 2	13.102127
3	Rocket League	10.198863
4	Counter-Strike: Global Offensive	9.425263
5	Apex Legends	8.024147
6	Rainbow Six Siege	5.566971
7	Call of Duty: Warzone	5.163371
8	PLAYERUNKNOWNâ□□S BATTLEGROUNDS	4.922998
9	VALORANT	3.932004
10	Halo Infinite	3.901842



	Game	Pct
1	League of Legends	31.999626
2	StarCraft II	19.100242
3	Overwatch	13.854852
4	PLAYERUNKNOWNâ□□S BATTLEGROUNDS	9.722204
5	Overwatch 2	5.717763
6	StarCraft: Brood War	5.220286
7	StarCraft: Remastered	4.526990
8	Heroes of the Storm	4.495270
9	Dota 2	3.019037
10	VALORANT	2.343730



Author Contribution

Isabella A.: GitHub repository setup, Quarto document setup, data analysis 1, Plan document. Adam R.: Presentation setup, GitHub contributions, data analysis 2, data review. #Code Appendix

```r
# Code by Isabella. Reviewed by Adam.
# Using the Google code formatting guidelines.
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.1
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(scales)
```

```
Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```r
earnings_raw <- tribble(
  ~rank, ~country, ~total_earnings, ~players,
  1, "China", "$333,124,853.95", 9387,
  2, "United States", "$302,289,323.81", 29501,
  3, "Korea, Republic of", "$157,154,934.92", 5987,
  4, "Russian Federation", "$98,801,881.46", 5900,
```

```r
  5, "Brazil", "$74,212,435.26", 6058,
  6, "Denmark", "$64,982,127.80", 2212,
  7, "France", "$63,290,937.28", 6705,
  8, "Sweden", "$59,203,302.47", 3367,
  9, "Germany", "$53,321,254.74", 6987,
  10, "Canada", "$50,919,206.73", 4395,
  11, "United Kingdom", "$50,674,463.56", 5747,
  12, "Japan", "$43,535,932.56", 4004,
  13, "Ukraine", "$41,266,588.03", 1727,
  14, "Australia", "$34,849,291.99", 4639,
  15, "Finland", "$34,242,302.63", 2216,
  16, "Poland", "$32,938,128.16", 3164,
  17, "Thailand", "$31,614,841.27", 2319,
  18, "Philippines", "$27,381,660.37", 1918,
  19, "Malaysia", "$23,483,101.29", 1723,
  20, "Taiwan, Republic of China", "$23,126,488.07", 1726
)

earnings_tidy <- earnings_raw %>%
  mutate(
    total_earnings = parse_number(total_earnings),
    players = as.integer(players),
    earnings_per_player = total_earnings / players
  ) %>%
  arrange(desc(total_earnings))

table_caption <- paste(
  "Table 1. Top 20 countries by total eSports earnings.",
  "Columns include rank, country, total earnings (USD), number of players,",
  "and earnings per player (USD)."
)

table_alt_text <- paste(
  "A table listing the top 20 countries by total eSports earnings.",
  "China is highest at about 333 million USD, followed by the United States",
  "at about 302 million USD, then Korea, Republic of at about 157 million USD.",
  "The table also shows players and earnings per player for each country."
)

earnings_table <- earnings_tidy %>%
  transmute(
    Rank = rank,
```

```r
    Country = country,
    `Total Earnings (USD)` = dollar(total_earnings, accuracy = 1),
    Players = comma(players),
    `Earnings per Player (USD)` = dollar(earnings_per_player, accuracy = 1)
  )

cat(table_caption, "\n")
```

```
Table 1. Top 20 countries by total eSports earnings. Columns include rank, country, total ea
```

```r
cat("Alt text: ", table_alt_text, "\n\n", sep = "")
```

```
Alt text: A table listing the top 20 countries by total eSports earnings. China is highest a
```

```r
print(earnings_table, n = Inf)
```

```
# A tibble: 20 x 5
    Rank Country           `Total Earnings (USD)` Players Earnings per Player ~1
   <dbl> <chr>             <chr>                  <chr>   <chr>
 1     1 China             $333,124,854           9,387   $35,488
 2     2 United States     $302,289,324           29,501  $10,247
 3     3 Korea, Republic ~ $157,154,935           5,987   $26,249
 4     4 Russian Federati~ $98,801,881            5,900   $16,746
 5     5 Brazil            $74,212,435            6,058   $12,250
 6     6 Denmark           $64,982,128            2,212   $29,377
 7     7 France            $63,290,937            6,705   $9,439
 8     8 Sweden            $59,203,302            3,367   $17,583
 9     9 Germany           $53,321,255            6,987   $7,631
10    10 Canada            $50,919,207            4,395   $11,586
11    11 United Kingdom    $50,674,464            5,747   $8,818
12    12 Japan             $43,535,933            4,004   $10,873
13    13 Ukraine           $41,266,588            1,727   $23,895
14    14 Australia         $34,849,292            4,639   $7,512
15    15 Finland           $34,242,303            2,216   $15,452
16    16 Poland            $32,938,128            3,164   $10,410
17    17 Thailand          $31,614,841            2,319   $13,633
18    18 Philippines       $27,381,660            1,918   $14,276
19    19 Malaysia          $23,483,101            1,723   $13,629
20    20 Taiwan, Republic~ $23,126,488            1,726   $13,399
# i abbreviated name: 1: `Earnings per Player (USD)`
```

```r
plot_caption <- paste(
  "Figure 1. Total eSports earnings by country for the top 20 earning countries",
  "(USD, shown in millions)."
)

earnings_plot <- ggplot(
  earnings_tidy,
  aes(x = reorder(country, total_earnings), y = total_earnings)
) +
  geom_col(fill = "#2C7FB8") +
  coord_flip() +
  scale_y_continuous(labels = label_dollar(scale = 1e-6, suffix = "M")) +
  labs(
    title = "Highest Earning Countries in eSports",
    x = NULL,
    y = "Total Earnings (USD, Millions)",
    caption = plot_caption
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(size = 11),
    axis.title = element_text(size = 9),
    axis.text = element_text(size = 8),
    plot.caption = element_text(size = 8)
  )

cat("\n", plot_caption, "\n\n", sep = "")
```

```
Figure 1. Total eSports earnings by country for the top 20 earning countries (USD, shown in m
```

```r
print(earnings_plot)
```

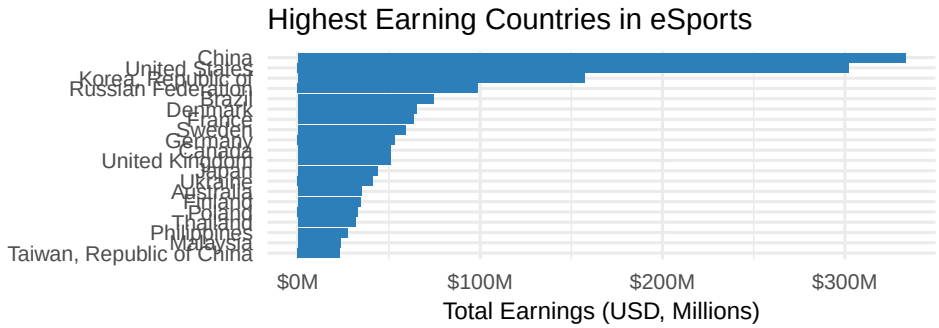


Figure 1. Total eSports earnings by country for the top 20 earning countries (USD, shown in millions).

Citation

"Esports Earnings: Prize Money / Results / History / Statistics." Esports Earnings, http://www.esportsearnings.com. Accessed 28 Apr. 2026.